

D654

Prototyping and Iterating I

Armored Drill

8 concepts · 12 items · 4 transfer items

What this is. Every concept in this course's public description, re-asked with more disguise than a practice test uses. Each stem carries facts that do not decide it. Every wrong answer is a named trap.

Source. Scope from the course's public description in the WGU Institutional Catalog (September 2026); items written from the field's standard body of knowledge. No WGU assessment item, course material or proprietary content is used or reproduced. Not affiliated with or endorsed by Western Governors University.

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How to run it

Paper first. Pencil. One pass, no notes, a timer if you want the pressure.

1. **Before you read the options**, fill the strip under the item: the target word (what is actually being asked), and the one fact in the stem that decides it.
2. **Principle before arithmetic**. Name the rule, then compute.
3. **Camouflage**. Most of every stem does not matter. Ask: which single fact decides this?
4. **Inverse pairs are the bait**. If two options are opposites, one of them is the trap built for you.
5. **Mark sure, unsure or guess** honestly. Sure-and-wrong is the most valuable miss on the page — it is a wrong mental model.
6. **Absolutes** (always, never, guarantee) are usually wrong. And recognizing an answer is not the same as owning it.
7. Items with a **heavy black bar** on the left are **transfer items**: the same principle, reversed or solved for a different variable.
8. Stop at the STOP page. Then grade, fill the triage grid, and tally the traps you fell for. The biggest tally is what you drill next.

The strip, filled in, looks like this:

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	_____	

Items

1.

A product designer at a meal-kit company with a new brand color palette has written six test tasks for a checkout mockup. The research budget is \$1,500. Her manager asks what the study is meant to learn, and she is not sure. What should she do first?

- A. Write research goals and questions, then revise the tasks so each one answers a goal.
- B. Add more tasks so that every screen is covered.
- C. Spend the \$1,500 on more participants to make the study stronger.
- D. Recruit participants now and set goals after seeing the results.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

2.

A bank's UX team plans a qualitative usability test of a new mobile deposit flow. The team has two weeks, a quiet conference room, and one user group. A vice president asks for 200 participants so the results are "statistically real." What is the best plan?

- A. Test with about five participants, fix the issues found, then run another small round.
- B. Test with 14 participants, one per day of the two-week window.
- C. Recruit all 200 participants for one round.
- D. Test with one participant, since any single user reveals all issues.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

3.

A streaming service's settings menu has a button labeled "Manage Profiles." The test lab was booked for Thursday afternoon. Which task wording is best for the usability test?

- A. "Find the setting."
- B. "Do you like the settings menu?"
- C. "Your younger brother is visiting and wants his own watch list. Set that up for him."
- D. "Click Manage Profiles and add a profile."

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

4. [TRANSFER — reversed stem / different variable]

In a test, all five participants completed a task in under ten seconds, but the task said “Go to Order History and find your last order.” What is the best conclusion?

- A.** The results do not show whether users can find order history on their own, because the task gave away the label.
- B.** A 100% success rate always means the design works.
- C.** Order history is easy to find; no changes are needed.
- D.** Five participants is too few to conclude anything.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

5.

A travel startup with a remote team across three time zones sees users abandon its trip-builder at the date step. The team wants to know why they quit, not just where. Which study type fits best?

- A.** An unmoderated test with many participants.
- B.** An analytics dashboard review only.
- C.** An unmoderated test, because the team is spread across time zones.
- D.** A moderated usability test, so a facilitator can ask follow-up questions as users struggle.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

6.

Twelve minutes into a session in a sunny corner office, a participant hovers over two buttons and asks the facilitator, “Is it this one?” The design lead, watching remotely, messages that the correct button is on the right. What should the facilitator do?

- A.** End the session, since the participant is confused.
- B.** Pass along the design lead’s message.
- C.** Point to the right button so the session stays on schedule.
- D.** Reply with a neutral prompt such as “What would you expect?” and let the participant choose.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

7. [TRANSFER — reversed stem / different variable]

A test report says every participant finished every task, yet support tickets about the same flow tripled after launch. Which facilitation problem best explains the gap?

- A.** The tasks were too realistic.
- B.** The facilitator gave hints or help during sessions, so struggles were never recorded.
- C.** The participants were asked to think aloud.
- D.** Support tickets are never related to usability.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

8.

After testing, a team has 140 sticky notes. The CEO attended one session and wants the logo made bigger. Four of five participants could not find the search bar, and one disliked the font. With one sprint to iterate, what should the team fix first?

- A.** The logo size.
- B.** The hidden search bar, found by 80% of participants.
- C.** The font, because every comment should be addressed.
- D.** Nothing until all 140 notes are fixed at once.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

9. [TRANSFER — reversed stem / different variable]

A team ranks issues by severity × frequency. Issue A blocks checkout for 2 of 5 users; issue B is a minor typo seen by 5 of 5. Which is higher priority and why?

- A.** They are tied, because both were seen by at least two users.
- B.** Issue B, because 100% of users saw it.
- C.** Neither; issues seen by fewer than all users are ignored.
- D.** Issue A, because a checkout blocker is severe even at 40% frequency.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

10.

A designer shows a polished, full-color checkout screen with final fonts and product photos from a recent photo shoot. Nothing on it can be clicked. A stakeholder asks to test whether users can move through checkout. What is the artifact, and is it ready for that test?

- A. A prototype; it is ready.
- B. A mockup; it is ready because the photos are final.
- C. A mockup; it needs links or interactions (a prototype) to test moving through checkout.
- D. A wireframe; add color before testing.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

11. [TRANSFER — reversed stem / different variable]

A team wants early feedback on page layout and content order before investing in visual design. Which artifact should it build?

- A. A high-fidelity mockup.
- B. A low-fidelity wireframe.
- C. A production release.
- D. A persona.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

12.

On a sign-up form built with the company’s award-winning design system, the “Create account” button is the same size and gray as “Cancel,” and field labels sit equally far from the field above and the field below. In testing, users typed their email into the password box. Which change best addresses both problems?

- A. Make both buttons bright blue.
- B. Color-code each label differently and leave spacing as is.
- C. Make “Create account” the visually dominant button and move each label closer to its own field.
- D. Keep the layout because the design system is award-winning.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

STOP

Answers and dissection follow. Finish every item and every strip first.

Key and dissection

Struck-through text is the camouflage: facts that do not decide the item.

1. Answer A Concept 1

A product designer at a meal-kit company ~~with a new brand color palette~~ has written six test tasks for a checkout mockup. ~~The research budget is \$1,500.~~ Her manager asks what the study is meant to learn, and she is not sure. What should she do first?

Target word: first step in a usability study plan **Decides:** Goals and research questions come before tasks.

A	Correct	Tasks exist to answer research questions; without goals, results cannot be judged.
B	Adjacent category	Coverage is not a goal; more tasks without purpose adds noise.
C	Camouflage	Budget does not fix a study with no stated purpose.
D	Inverted operation	Reverses the plan; goals shape who to recruit and what to measure.

2. Answer A Concept 2

A bank's UX team plans a qualitative usability test of a new mobile deposit flow. The team has two weeks, ~~a quiet conference room~~, and one user group. A vice president asks for 200 participants so the results are "statistically real." What is the best plan?

Target word: participant count for qualitative testing **Decides:** About five participants per group finds most major issues; iterate in rounds.

A	Correct	Small rounds catch most major problems; iterating finds more than one big round.
B	Wrong stem number	Builds the plan from the schedule instead of the research practice.
C	Adjacent category	Applies survey-style sampling to a qualitative test; costly and slow with diminishing returns.
D	Absolute language	One person misses many problems.

3. Answer C Concept 3

A streaming service's settings menu has a button labeled "Manage Profiles." ~~The test lab was booked for Thursday afternoon.~~ Which task wording is best for the usability test?

Target word: best task wording **Decides:** Realistic goal without the UI label.

A	Dropped element	Too vague; drops the goal the user is trying to reach.
B	Adjacent category	An opinion question, not a task.
C	Correct	Realistic scenario in user terms; does not reveal the button name.
D	Inverted operation	Gives the path, so it tests reading, not findability.

4. Answer A Concept 3 • transfer

In a test, all five participants completed a task in under ten seconds, but the task said "Go to Order History and find your last order." What is the best conclusion?

Target word: interpret results from a flawed task **Decides:** The task named the label, so success says little about findability.

A	Correct	Leading tasks inflate success.
B	Absolute language	Success depends on how the task was written.
C	Inverted operation	Draws the opposite conclusion from a flawed task.
D	Confusion set	Sample size is not the flaw here; the wording is.

5. Answer D Concept 4

A travel startup ~~with a remote team across three time zones~~ sees users abandon its trip-builder at the date step. The team wants to know why they quit, not just where. Which study type fits best?

Target word: moderated vs. unmoderated **Decides:** “Why” needs a facilitator who can probe.

A	Confusion set	Good for scale and where, weak for why.
B	Dropped element	Analytics show where users drop, which the team already knows.
C	Camouflage	Remote moderated sessions work fine across time zones.
D	Correct	Moderated sessions let the facilitator explore reasons.

6. Answer D Concept 5

Twelve minutes into a session ~~in a sunny corner office~~, a participant hovers over two buttons and asks the facilitator, “Is it this one?” The design lead, watching remotely, messages that the correct button is on the right. What should the facilitator do?

Target word: facilitator behavior **Decides:** Stay neutral; redirect and let the user decide.

A	Absolute language	Confusion is data, not a reason to stop.
B	Camouflage	The observer’s input does not change facilitator rules.
C	Inverted operation	Helping hides the very problem the test should find.
D	Correct	Neutral prompts keep the test honest and capture the confusion.

7. Answer B Concept 5 • transfer

A test report says every participant finished every task, yet support tickets about the same flow tripled after launch. Which facilitation problem best explains the gap?

Target word: diagnose facilitation from outcome **Decides:** Facilitator help masks problems.

A	Adjacent category	Realistic tasks are good practice.
B	Correct	Assisted completion looks like success in the report.
C	Inverted operation	Think-aloud surfaces problems; it does not hide them.
D	Absolute language	Tickets are a common usability signal.

8. Answer B Concept 6

After testing, a team has 140 sticky notes. ~~The CEO attended one session and wants the logo made bigger.~~ Four of five participants could not find the search bar, and one disliked the font. With one sprint to iterate, what should the team fix first?

Target word: prioritize a finding **Decides:** Severity and frequency: 4 of 5 blocked on search.

A	Camouflage	A stakeholder preference, not a test finding.
B	Correct	High frequency (4/5 = 80%) and blocks a task, so highest priority.
C	Absolute language	One mild preference ranks below a task-blocking issue.
D	Dropped element	Skips prioritization.

9. Answer D Concept 6 · transfer

A team ranks issues by severity × frequency. Issue A blocks checkout for 2 of 5 users; issue B is a minor typo seen by 5 of 5. Which is higher priority and why?

Target word: severity outweighs frequency for blocking issues **Decides:** A blocks a critical task.

A	Wrong stem number	Uses the counts without the severity.
B	Dropped element	Ignores severity.
C	Absolute language	No such rule exists.
D	Correct	Severity can outweigh frequency; a blocked purchase is critical.

10. Answer C Concept 7

A designer shows a polished, full-color checkout screen with final fonts and product photos ~~from a recent photo shoot~~. Nothing on it can be clicked. A stakeholder asks to test whether users can move through checkout. What is the artifact, and is it ready for that test?

Target word: name the fidelity level and fit **Decides:** Static and high-fidelity = mockup; flow testing needs interaction.

A	Confusion set	Polish does not make it interactive.
B	Camouflage	Final photos do not make the screen clickable.
C	Correct	Mockups are static visuals; flow testing needs a clickable prototype.
D	Inverted operation	It already has full visual design; the gap is interaction.

11. Answer B Concept 7 · transfer

A team wants early feedback on page layout and content order before investing in visual design. Which artifact should it build?

Target word: choose fidelity for the question **Decides:** Structure questions call for a wireframe.

A	Inverted operation	Spends effort on visuals before structure is settled.
B	Correct	Wireframes test structure cheaply before visual polish.
C	Adjacent category	Far too late and costly for early layout feedback.
D	Confusion set	A research artifact, not a design artifact.

12. Answer C Concept 8

On a sign-up form ~~built with the company's award-winning design system~~, the "Create account" button is the same size and gray as "Cancel," and field labels sit equally far from the field above and the field below. In testing, users typed their email into the password box. Which change best addresses both problems?

Target word: visual hierarchy and proximity **Decides:** Primary action needs emphasis; labels must sit close to their own field.

A	Dropped element	Equal emphasis again, and the label spacing is untouched.
B	Adjacent category	Relies on color instead of proximity to show grouping.
C	Correct	Contrast marks the primary action; proximity pairs labels with fields.
D	Camouflage	Testing shows a problem; the system's reputation does not decide it.

Triage grid

Write each item number in the box it belongs to.

	Right	Wrong
Sure	Owned.	Concept inversion — top priority. A wrong mental model; it survives rewording.
Unsure / guess	Recognition, not mastery. Re-drill with the transfer.	Not learned yet.

Trap tally

For every miss, find the letter you chose in the key and tally its trap. The biggest number is the pattern to drill.

Trap	What it does	In this drill	Your misses
Dropped element	Leaves out a required term or step.	5	
Wrong stem number	A figure copied from the prompt that is not the answer.	2	
Adjacent category	The right idea in the wrong bucket.	6	
Inverted operation	Direction flipped: add/subtract, cause/effect, higher/lower.	7	
Absolute language	Always, never, guarantee — where judgment applies.	6	
Confusion set	Every option is real; the test is exact role-matching.	4	
Camouflage	Answers a stem fact that was never the question.	6	

Anchor rows — the concept list

One row per concept: the rule to lock in, the trap that catches people, and the tell in the question that signals it.

#	Concept	Rule	Trap	Tell
1	Research goals drive the study plan	A usability study plan starts with research goals and questions; tasks, participants, and metrics follow from them.	Writing tasks first and deciding what to learn afterward.	A plan with tasks but no statement of what the team is trying to learn.
2	Small samples find most usability issues	About five participants per user group uncover most major usability problems in a qualitative study; run more rounds rather than more people.	Believing a usability test needs hundreds of participants to be valid.	A budget or schedule question about how many participants to recruit.
3	Realistic, non-leading tasks	Tasks describe a realistic goal in the user's words without naming the button or path to use.	Writing tasks that use the interface labels, which tell the user where to click.	A task that repeats the exact menu or button text.
4	Moderated vs. unmoderated testing	Moderated tests let a facilitator probe and follow up; unmoderated tests scale cheaply but cannot ask why in the moment.	Choosing unmoderated when the team needs to explore reasons behind behavior.	"We need to understand why" vs. "We need many sessions fast."
5	Facilitator neutrality and think-aloud	The facilitator asks participants to think aloud and does not help, hint, or defend the design.	Rescuing a struggling participant, which hides the problem.	A participant asks the facilitator, "Is this right?"
6	Synthesizing and prioritizing findings	Group observations (affinity diagram) into themes and prioritize by severity and frequency before iterating.	Fixing whatever the loudest participant or stakeholder mentioned first.	A long list of raw notes and a limited sprint.
7	Fidelity: wireframe, mockup, prototype	Wireframes show structure; mockups are static, high-fidelity visuals; prototypes add interaction.	Calling any polished screen a prototype, or testing flow on a static mockup.	Whether the artifact is clickable and how finished it looks.
8	Visual hierarchy and Gestalt	Size, contrast, spacing, and proximity signal importance and grouping; related items sit close together, and the primary action stands out.	Giving every element equal weight, or grouping by color alone.	Users miss the main action or pair a label with the wrong field.

Course scope (public description): Prototyping and Iterating I focuses on usability testing to create user-centered digital products. Tools and techniques are explored, including planning and conducting a usability study and iterating on a mockup that incorporates visual design principles. Students will collaborate to gather feedback to apply to a mockup for continuous improvement. This course provides hands-on application for producing industry artifacts for the student's professional portfolio.